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## Information Architecture Document

### AI, Technology, Research & Venture Intelligence Dashboard

#### Deliverable 2 — Full Platform Information Architecture

**Document type:** Information Architecture Blueprint (no UI, no code, no schema, no API spec) **Audience:** Product design, frontend engineering, backend engineering, AI engineering **Status:** Pre-design, pre-development reference architecture

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## SECTION 1 — EXECUTIVE INFORMATION ARCHITECTURE

### 1.1 Platform Purpose

The platform is a **Personal AI & Market Intelligence System** — a single surface that continuously ingests, structures, scores, and connects information from the AI, research, venture, policy, and open-source ecosystems, and turns it into decisions rather than headlines.

It exists to answer six standing questions on every piece of information it touches:

| Question                            | System Responsibility             |
|-------------------------------------|-----------------------------------|
| What happened?                      | Ingestion + summarization         |
| Why does it matter?                 | Scoring + context linking         |
| Does it affect my businesses?       | Business Relevance System         |
| Should I take action?               | Priority classification + alerts  |
| Can I search previous intelligence? | Search Architecture               |
| Can I compare trends over time?     | Historical database + trend views |

The platform is explicitly **not** a news reader or aggregator. A news reader terminates at “here is an article.” This system terminates at “here is what changed, why it matters to you specifically, and what you can do next,” with every item permanently stored, cross-linked, and retrievable.

### 1.2 User Goals

Primary user (V1): a single power user — founder/operator — who needs a daily edge in AI, research, and venture markets across two businesses (Viorant, Miraya) plus personal investing/positioning interests.

| Goal              | Description                                                                            |
|-------------------|----------------------------------------------------------------------------------------|
| Stay current      | Absorb the day’s AI/VC/research signal in minutes, not hours                           |
| Filter noise      | See only what is credible, fresh, and relevant                                         |
| Understand impact | Know immediately if an item touches an existing business, competitor, or opportunity   |
| Retain knowledge  | Never lose a useful item; everything is searchable forever                             |
| Track trends      | See how a topic, company, or technology evolves over weeks/months                      |
| Act quickly       | Convert intelligence into notes, bookmarks, or follow-ups without leaving the platform |

### 1.3 Primary Workflows

The IA is organized around five recurring behavioral loops, detailed fully in Section 7:

1. **Morning Brief** — daily top-down scan of everything new and important
2. **Deep Research** — vertical drill-down into one topic/company/paper
3. **Discovery** — horizontal exploration of a category (e.g., funding, models)
4. **Retrieval** — search/recall of something seen before
5. **Curation** — save, tag, annotate, and organize for later use or sharing

## 1.4 Information Hierarchy

The platform’s information is organized in a strict seven-level hierarchy, applied consistently to every module (fully diagrammed in Section 17):

Platform → Module → Category → Subcategory → Object → Detail Page → Related Objects/Actions

This hierarchy is the single organizing law of the IA: every module, no matter how different its content (a research paper vs. a funding round vs. a GitHub repo), resolves down to an **Object** with a **Detail Page**, surrounded by **Related Objects** and **Actions**. This uniformity is what allows one design system and one backend data model to serve twelve+ content domains.

## 1.5 Navigation Philosophy

Five principles govern navigation across the entire platform:

| Principle                                        | Meaning                                                                                                                                 |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Flat before deep</b>                          | Every module is reachable in one click from the global sidebar; depth is reserved for object detail, not for finding the module itself  |
| <b>Consistent module anatomy</b>                 | Every content module (News, Funding, Research, etc.) has the same internal shape: Feed → Filters/Sort → Detail Page → Related → Actions |
| <b>Search is a parallel path, not a fallback</b> | Global search is always available and is treated as a first-class navigation method, not a last resort                                  |
| <b>Context follows the user</b>                  | Filters, scroll position, and active tab persist when the user navigates to a detail page and returns                                   |
| <b>One home for saved things</b>                 | Bookmarks, notes, reading list, and saved searches live under a single “Saved Intelligence” area, never scattered per-module            |

## 1.6 Design Principles

| Principle                                 | Application                                                                                                                              |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Signal over volume</b>                 | Every list defaults to a scored/ranked view, never raw chronological firehose, though chronological is always available as a sort option |
| <b>Business context is never optional</b> | Business Relevance is a first-class field on every object, not a bolt-on filter                                                          |

| Principle                                   | Application                                                                                                                                                                                        |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Everything is an object</b>              | News, papers, companies, people, repos are all modeled as objects with IDs, so relationships and search work uniformly                                                                             |
| <b>Progressive disclosure</b>               | Feed cards show summary + score; detail pages show full depth; nothing is hidden behind more than one extra click                                                                                  |
| <b>Local-first cost discipline</b>          | Given the near-₹0 budget constraint, the architecture favors a single relational store, scheduled batch ingestion, and on-demand AI calls over always-on infrastructure (elaborated in Section 20) |
| <b>Designed for multi-user from day one</b> | Every object and preference is modeled as belonging to a <code>user_id</code> scope even though V1 ships single-user, so multi-user is a config change, not a rebuild                              |

## SECTION 2 — GLOBAL NAVIGATION STRUCTURE

The global navigation is a persistent left sidebar (desktop) / bottom nav + drawer (mobile). It is organized into five zones: **Overview**, **Intelligence Modules**, **Personal**, **System**, and **Future** (visually hidden/flagged until built).

### 2.1 Navigation Zones

#### ZONE A — OVERVIEW

- Dashboard

#### ZONE B — INTELLIGENCE MODULES

- AI News
- Models & Releases
- Funding
- VC Intelligence
- Research
- Universities
- Government
- Open Source
- Developer Ecosystem
- Social Intelligence
- Business Intelligence

#### ZONE C — PERSONAL

- Saved Intelligence
  - Bookmarks
  - Reading List
  - Notes
  - Collections

#### ZONE D — SYSTEM

- Settings
- Profile

Admin

ZONE E – FUTURE (flagged, hidden by default)

AI Chat

Trend Prediction

Team Workspace

## 2.2 Module Reference Table

| Module                       | Purpose                                                                            | Target User                          | Importance  | Expected Frequency                  |
|------------------------------|------------------------------------------------------------------------------------|--------------------------------------|-------------|-------------------------------------|
| <b>Dashboard</b>             | Single-pane daily overview: top stories, alerts, scores, trends across all modules | All users                            | Critical    | Every session (daily, multiple/day) |
| <b>AI News</b>               | Curated, scored feed of AI industry news                                           | All users                            | Critical    | Daily                               |
| <b>Models &amp; Releases</b> | Tracks new model launches, benchmarks, versioning across labs                      | Builders, technical evaluators       | High        | Weekly, spikes on release days      |
| <b>Funding</b>               | Startup funding rounds — amount, stage, investors, sector                          | Investors, founders, market watchers | Critical    | Daily                               |
| <b>VC Intelligence</b>       | Investor/fund-level view: thesis, portfolio, recent activity                       | Investors, fundraising founders      | High        | Weekly                              |
| <b>Research</b>              | Academic papers, preprints, benchmarks                                             | Technical/research-minded users      | High        | Weekly                              |
| <b>Universities</b>          | Lab-level and institution-level research output tracking                           | Research-focused users               | Medium      | Monthly                             |
| <b>Government</b>            | Policy, regulation, funding programs, export controls                              | Compliance/strategy-minded users     | Medium-High | Weekly, spikes on policy events     |
| <b>Open Source</b>           | Notable repos, framework releases, community traction                              | Engineers, technical evaluators      | High        | Weekly                              |
| <b>Developer Ecosystem</b>   | Tooling, SDKs, platforms, dev-facing product launches                              | Engineers, product builders          | Medium      | Weekly                              |
| <b>Social Intelligence</b>   | Signal from X/LinkedIn/HN/Reddit — sentiment, virality,                            | All users, especially trend-watchers | Medium      | Daily                               |

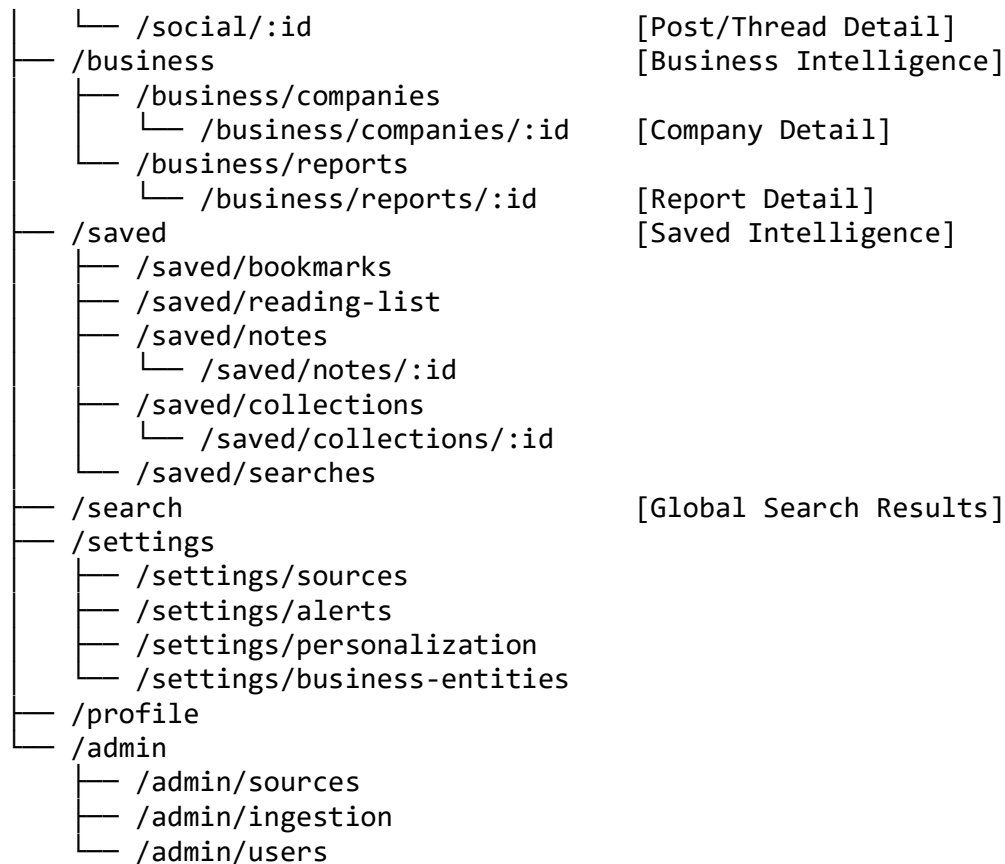


| Module                       | Purpose                                                                  | Target User          | Importance | Expected Frequency |
|------------------------------|--------------------------------------------------------------------------|----------------------|------------|--------------------|
|                              | discourse                                                                |                      |            |                    |
| <b>Business Intelligence</b> | Competitor tracking, market sizing, sector reports                       | Founders/operators   | Critical   | Weekly             |
| <b>Saved Intelligence</b>    | Bookmarks, reading list, notes, collections                              | All users            | High       | Daily              |
| <b>Settings</b>              | Preferences, sources, alerts, personalization controls                   | All users            | Medium     | Occasional         |
| <b>Profile</b>               | Account, business entities, interests                                    | All users            | Low-Medium | Occasional         |
| <b>Admin</b>                 | Source management, ingestion health, user management (multi-user future) | Platform owner/admin | Medium     | Occasional         |

## SECTION 3 — COMPLETE PAGE HIERARCHY

### 3.1 Sitemap Structure (textual)

|                         |                         |
|-------------------------|-------------------------|
| / (Dashboard)           |                         |
| ├ /news                 | [AI News]               |
| │ └ /news/:id           | [News Detail]           |
| ├ /models               | [Models & Releases]     |
| │ └ /models/:id         | [Model Detail]          |
| ├ /funding              | [Funding]               |
| │ └ /funding/:id        | [Funding Round Detail]  |
| ├ /vc                   | [VC Intelligence]       |
| │ └ /vc/investors       | [Investor List]         |
| │ │ └ /vc/investors/:id | [Investor Detail]       |
| │ └ /vc/funds           | [Fund List]             |
| │ │ └ /vc/funds/:id     | [Fund Detail]           |
| ├ /research             | [Research]              |
| │ └ /research/:id       | [Paper Detail]          |
| ├ /universities         | [Universities]          |
| │ └ /universities/:id   | [Institution Detail]    |
| ├ /government           | [Government]            |
| │ └ /government/:id     | [Policy Detail]         |
| ├ /opensource           | [Open Source]           |
| │ └ /opensource/:id     | [Repository Detail]     |
| ├ /developer            | [Developer Ecosystem]   |
| │ └ /developer/:id      | [Tool/Framework Detail] |
| └ /social               | [Social Intelligence]   |



### 3.2 Page Definition Table (representative — pattern repeats per module)

| Page                  | Parent        | Children                 | Breadcrumb                         | Relationships                                           |
|-----------------------|---------------|--------------------------|------------------------------------|---------------------------------------------------------|
| Dashboard             | —             | none (root)              | Home                               | Pulls top items from every module                       |
| AI News (feed)        | Dashboard     | News Detail              | Home / AI News                     | Links to Companies, Models, People mentioned            |
| News Detail           | AI News       | none                     | Home / AI News / [Headline]        | Related News, Related Companies, Related Research       |
| Funding (feed)        | Dashboard     | Funding Round Detail     | Home / Funding                     | Links to Startups, Investors, Sectors                   |
| Funding Round Detail  | Funding       | none                     | Home / Funding / [Company – Round] | Startup, Investors, prior/future rounds of same company |
| VC Intelligence (hub) | Dashboard     | Investor List, Fund List | Home / VC Intelligence             | —                                                       |
| Investor              | Investor List | none                     | Home / VC                          | Portfolio                                               |

| Page                     | Parent                                     | Children                                    | Breadcrumb                           | Relationships                                            |
|--------------------------|--------------------------------------------|---------------------------------------------|--------------------------------------|----------------------------------------------------------|
| Detail                   |                                            |                                             | Intelligence / Investors / [Name]    | companies, co-investors, fund                            |
| Research (feed)          | Dashboard                                  | Paper Detail                                | Home / Research                      | Authors, Institutions, cited Companies                   |
| Paper Detail             | Research                                   | none                                        | Home / Research / [Title]            | Authors, Institution, related papers, spun-out companies |
| Company Detail           | Business Intelligence / Funding / Research | none                                        | Home / Business / Companies / [Name] | Founders, funding history, research lineage, competitors |
| Saved Intelligence (hub) | Dashboard                                  | Bookmarks, Reading List, Notes, Collections | Home / Saved                         | Pulls from every module                                  |

**Navigation Flow rule:** every list page → detail page transition preserves scroll position and active filters on back-navigation. Every detail page surfaces a “Related Objects” rail that cross-links to other modules (this is what separates the IA from a flat CMS).

## SECTION 4 — MODULE BREAKDOWN

Each module below follows the same ten-part definition so that engineering can treat modules as configurations of one system rather than twelve bespoke builds.

### 4.1 AI News

| Attribute             | Detail                                                                                          |
|-----------------------|-------------------------------------------------------------------------------------------------|
| Purpose               | Surface and score daily AI industry news                                                        |
| Primary Data Sources  | Publisher RSS/APIs, company blogs, aggregator APIs, curated newsletters                         |
| Displayed Information | Headline, source, summary (AI-generated), score, tags, published time, business relevance badge |
| Filters               | Date, Source, Topic/Technology, Company, Priority, Business Relevance, Read/Unread              |
| Sorting               | Newest, Most Important, Trending, Business Relevance                                            |
| Ranking               | Intelligence Score (Section 11)                                                                 |

| Attribute           | Detail                                                                                       |
|---------------------|----------------------------------------------------------------------------------------------|
| Actions             | Bookmark, Add to Reading List, Note, Tag, Mark Business-Relevant, Share/Export               |
| Search Capabilities | Keyword, natural language, filtered search within module                                     |
| Future Expansion    | Auto-clustering of duplicate coverage across sources; AI-generated “why it matters” per item |
| Dependencies        | Company object, Model object, Person object (for entity linking)                             |

#### 4.2 Models & Releases

| Attribute             | Detail                                                                         |
|-----------------------|--------------------------------------------------------------------------------|
| Purpose               | Track new model launches, versions, and benchmark movement                     |
| Primary Data Sources  | Lab announcement pages, Hugging Face, benchmark leaderboards, arXiv            |
| Displayed Information | Model name, lab, modality, benchmark deltas, release notes summary, license    |
| Filters               | Lab, Modality, License Type, Date, Benchmark Category                          |
| Sorting               | Newest, Highest Benchmark Gain, Most Discussed                                 |
| Ranking               | Technical Innovation Score + Community Engagement                              |
| Actions               | Bookmark, Compare Models, Note                                                 |
| Search Capabilities   | Keyword + filter by lab/modality                                               |
| Future Expansion      | Side-by-side model comparison tool; benchmark trend charts over time           |
| Dependencies          | Company (lab) object, Research Paper object (if model has an associated paper) |

#### 4.3 Funding

| Attribute             | Detail                                                                       |
|-----------------------|------------------------------------------------------------------------------|
| Purpose               | Track startup funding rounds                                                 |
| Primary Data Sources  | Funding databases/APIs, press releases, VC blog posts                        |
| Displayed Information | Company, round stage, amount, lead/participating investors, sector, date     |
| Filters               | Stage, Sector, Country, Investor, Amount Range, Date                         |
| Sorting               | Newest, Highest Funding, Most Investors, Business Relevance                  |
| Ranking               | Funding Score + Business Impact                                              |
| Actions               | Bookmark, Add to Watchlist, Note, Mark Competitor                            |
| Search Capabilities   | Keyword, filtered, natural language (“seed rounds in agentic AI this month”) |
| Future Expansion      | Cap table / valuation trend estimation; auto-competitor flagging             |

| Attribute    | Detail                                          |
|--------------|-------------------------------------------------|
| Dependencies | Startup object, Investor object, Founder object |

#### 4.4 VC Intelligence

| Attribute             | Detail                                                                            |
|-----------------------|-----------------------------------------------------------------------------------|
| Purpose               | Investor and fund-level intelligence — thesis, activity, portfolio                |
| Primary Data Sources  | Fund websites, portfolio pages, funding data cross-referenced from Funding module |
| Displayed Information | Fund/investor name, thesis tags, stage focus, recent investments, portfolio size  |
| Filters               | Fund Size, Stage Focus, Sector Focus, Region, Activity Recency                    |
| Sorting               | Most Active, Newest Investment, Portfolio Size                                    |
| Ranking               | Strategic Value + Activity Score                                                  |
| Actions               | Bookmark, Follow Investor, Note                                                   |
| Search Capabilities   | Keyword, filter by thesis/sector                                                  |
| Future Expansion      | Investor CRM (Section 16), co-investment network graph                            |
| Dependencies          | Investor object, Fund object, Funding object                                      |

#### 4.5 Research

| Attribute             | Detail                                                                  |
|-----------------------|-------------------------------------------------------------------------|
| Purpose               | Surface relevant academic papers/preprints                              |
| Primary Data Sources  | arXiv, Semantic Scholar, Papers with Code, conference proceedings       |
| Displayed Information | Title, authors, institution, abstract summary, citation count, category |
| Filters               | Category, Institution, Author, Date, Citation Count Range               |
| Sorting               | Newest, Most Cited, Most Important (score), Trending                    |
| Ranking               | Research Importance Score                                               |
| Actions               | Bookmark, Add to Reading List, Note, Highlight                          |
| Search Capabilities   | Keyword, natural language, semantic (future)                            |
| Future Expansion      | Vector/semantic search, citation graph view                             |
| Dependencies          | Author (Person) object, Institution object, Company (if spun out)       |

#### 4.6 Universities

| Attribute             | Detail                                                          |
|-----------------------|-----------------------------------------------------------------|
| Purpose               | Track institution-level and lab-level research output           |
| Primary Data Sources  | University lab pages, department feeds, research aggregators    |
| Displayed Information | Institution name, notable labs, recent papers, notable spinouts |
| Filters               | Region, Research Focus Area, Notable Output Recency             |
| Sorting               | Most Active, Most Cited Output                                  |

| Attribute           | Detail                                             |
|---------------------|----------------------------------------------------|
| Ranking             | Research Importance aggregate at institution level |
| Actions             | Bookmark, Follow Institution                       |
| Search Capabilities | Keyword, filter by focus area                      |
| Future Expansion    | Lab-level sub-tracking, faculty movement tracking  |
| Dependencies        | Research Paper object, Person (Author) object      |

#### 4.7 Government

| Attribute             | Detail                                                                      |
|-----------------------|-----------------------------------------------------------------------------|
| Purpose               | Track policy, regulation, and public funding relevant to AI/tech            |
| Primary Data Sources  | Government portals, regulatory bodies, official press releases              |
| Displayed Information | Policy title, jurisdiction, summary, status (proposed/enacted), impact area |
| Filters               | Jurisdiction/Country, Policy Type, Status, Impact Area                      |
| Sorting               | Newest, Highest Impact, Status                                              |
| Ranking               | Business Impact + Strategic Value                                           |
| Actions               | Bookmark, Note, Mark Business-Relevant                                      |
| Search Capabilities   | Keyword, filtered                                                           |
| Future Expansion      | Regulatory timeline tracker, jurisdiction comparison view                   |
| Dependencies          | Company object (if policy names specific companies/sectors)                 |

#### 4.8 Open Source

| Attribute             | Detail                                                               |
|-----------------------|----------------------------------------------------------------------|
| Purpose               | Track notable repositories and OSS framework activity                |
| Primary Data Sources  | GitHub API/trending, package registries                              |
| Displayed Information | Repo name, org, stars/growth, description, license, primary language |
| Filters               | Language, License, Category, Star Growth Rate, Date                  |
| Sorting               | Trending, Most Stars, Newest, Fastest Growing                        |
| Ranking               | Community Engagement + Technical Innovation                          |
| Actions               | Bookmark, Note, Star-Track (watch growth)                            |
| Search Capabilities   | Keyword, filtered                                                    |
| Future Expansion      | Growth trend charts, dependency graph view                           |
| Dependencies          | Company/Org object, Developer Tool object                            |

#### 4.9 Developer Ecosystem

| Attribute | Detail                                            |
|-----------|---------------------------------------------------|
| Purpose   | Track developer-facing tools, SDKs, and platforms |

| Attribute             | Detail                                                           |
|-----------------------|------------------------------------------------------------------|
| Primary Data Sources  | Product Hunt, dev blogs, changelogs, platform announcement pages |
| Displayed Information | Tool name, category, publisher, key features, pricing model      |
| Filters               | Category, Pricing Model, Publisher, Date                         |
| Sorting               | Newest, Most Discussed, Business Relevance                       |
| Ranking               | Technical Innovation + Community Engagement                      |
| Actions               | Bookmark, Note, Mark Evaluate                                    |
| Search Capabilities   | Keyword, filtered                                                |
| Future Expansion      | Tool comparison matrix                                           |
| Dependencies          | Company object, Open Source object (if tool is OSS)              |

#### 4.10 Social Intelligence

| Attribute             | Detail                                                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------|
| Purpose               | Surface high-signal discourse and sentiment from social platforms                                    |
| Primary Data Sources  | X/Twitter API, LinkedIn, Hacker News, Reddit                                                         |
| Displayed Information | Post excerpt (paraphrased/summarized, not reproduced), author, platform, engagement stats, sentiment |
| Filters               | Platform, Topic, Sentiment, Author, Date                                                             |
| Sorting               | Most Discussed, Trending, Newest                                                                     |
| Ranking               | Community Engagement Score                                                                           |
| Actions               | Bookmark, Note                                                                                       |
| Search Capabilities   | Keyword, filtered                                                                                    |
| Future Expansion      | Sentiment trend charts, influencer tracking                                                          |
| Dependencies          | Person object, Company object (entity mentions)                                                      |

#### 4.11 Business Intelligence

| Attribute             | Detail                                                                                |
|-----------------------|---------------------------------------------------------------------------------------|
| Purpose               | Competitor tracking, market sizing, sector reports, strategic view for Viorant/Miraya |
| Primary Data Sources  | Aggregated from all other modules + manual curation + market reports                  |
| Displayed Information | Company profiles, competitor moves, sector summaries, opportunity/risk flags          |
| Filters               | Business Entity (Viorant/Miraya), Category, Competitor, Opportunity Type              |

| Attribute           | Detail                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| Sorting             | Highest Business Relevance, Newest, Priority                                                                         |
| Ranking             | Business Relevance Score (Section 12)                                                                                |
| Actions             | Bookmark, Note, Add to Competitor Watchlist, Flag Opportunity                                                        |
| Search Capabilities | Keyword, filtered, natural language                                                                                  |
| Future Expansion    | Competitive intelligence dashboards, market forecasting                                                              |
| Dependencies        | Company, Startup, Funding, News, Research objects (this module is largely a relevance-filtered rollup of all others) |

## SECTION 5 — DATA OBJECTS

Every content item on the platform is modeled as one of the object types below. All objects share a **Common Object Envelope**, then extend it with type-specific fields.

### 5.1 Common Object Envelope (applies to every object type)

| Field Group        | Fields                                                                        |
|--------------------|-------------------------------------------------------------------------------|
| Identity           | id (unique identifier), type (object type), title/name                        |
| Metadata           | source, source_url, source_credibility, ingested_at, published_at, updated_at |
| Status             | status (new / reviewed / archived), read_status (read/unread)                 |
| Tags               | topic_tags[], technology_tags[], region_tags[]                                |
| Business Relevance | relevance_score, relevant_entities[] (Viorant/Miraya/None), relevance_reason  |
| Scoring            | intelligence_score, priority_class (Critical/High/Medium/Low/Noise)           |
| Relationships      | related_object_ids[] (typed references to other objects)                      |
| Timestamps         | created_at, updated_at, last_viewed_at                                        |

### 5.2 Object Type Definitions

#### News

- **Fields:** headline, summary (AI-generated), body\_excerpt, source, author, publish\_date, category
- **Relationships:** mentions Companies, Models, People, Research Papers
- **Business Relevance:** yes — direct field

#### Research Paper

- **Fields:** title, abstract\_summary, authors[], institution, publish\_date, category, citation\_count, arxiv\_id/doi



- **Relationships:** Authors (Person), Institution, cited/citing papers, spun-out Company
- **Business Relevance:** derived from topic match to business focus areas

### Company

- **Fields:** name, description, sector, founding\_date, HQ location, website, status (active/acquired/shut down)
- **Relationships:** Founders (Person), Funding rounds, Investors, Products, competitors
- **Business Relevance:** direct field (competitor / partner / customer opportunity / none)

### Startup

- **Fields:** (extends Company) stage, total\_raised, last\_round\_date, sector
- **Relationships:** Founders, Investors, Funding rounds
- **Business Relevance:** direct field

### Investor

- **Fields:** name, type (individual/angel), focus\_areas[], notable\_investments[]
- **Relationships:** Portfolio Companies, Fund (if affiliated), co-investors
- **Business Relevance:** relevant if investing in adjacent space or a fundraising target

### VC Fund

- **Fields:** fund\_name, fund\_size, stage\_focus[], sector\_focus[], founded\_year
- **Relationships:** Investors (partners), Portfolio Companies
- **Business Relevance:** relevant as fundraising target

### Founder

- **Fields:** (extends Person) current\_company, past\_companies[], bio\_summary
- **Relationships:** Company, Institution (education), co-founders
- **Business Relevance:** relevant if founder of competitor/partner

### Research Institution

- **Fields:** name, country, notable\_labs[], focus\_areas[]
- **Relationships:** Papers, Authors, Spinout Companies
- **Business Relevance:** relevant if output overlaps business focus areas

### Government Policy

- **Fields:** title, jurisdiction, policy\_type, status, summary, effective\_date, impact\_area[]
- **Relationships:** affected Companies/Sectors
- **Business Relevance:** direct field — regulatory exposure

### Repository (Open Source)

- **Fields:** repo\_name, org, description, language, license, stars, star\_growth\_rate

- **Relationships:** Org (Company), Contributors (Person), Developer Tool
- **Business Relevance:** relevant if in stack or competitive tooling space

#### AI Model

- **Fields:** model\_name, lab (Company), modality, parameters (if disclosed), benchmark\_scores{}, license, release\_date
- **Relationships:** Lab, associated Research Paper
- **Business Relevance:** relevant if usable in or competitive with product stack

#### Framework

- **Fields:** name, publisher, category, description, adoption\_signal
- **Relationships:** Publisher (Company), related Repositories
- **Business Relevance:** relevant if part of tech stack decisions

#### Developer Tool

- **Fields:** name, publisher, category, pricing\_model, description
- **Relationships:** Publisher (Company), related Frameworks
- **Business Relevance:** relevant to product/eng tooling decisions

#### Social Post

- **Fields:** platform, author (Person), summary (paraphrased), engagement\_stats{}, sentiment, posted\_at
- **Relationships:** mentions Companies/People/Topics
- **Business Relevance:** derived from mention relevance

#### Event

- **Fields:** event\_name, event\_type (conference/launch/hearing), date, location, description
- **Relationships:** related Companies, People, Policies
- **Business Relevance:** direct field

### 5.3 Object Field Summary Matrix

| Object         | Unique ID | Core Fields Count | Has Business Relevance | Has Timestamps | Has Tags |
|----------------|-----------|-------------------|------------------------|----------------|----------|
| News           | ✓         | 7                 | ✓                      | ✓              | ✓        |
| Research Paper | ✓         | 8                 | ✓ (derived)            | ✓              | ✓        |
| Company        | ✓         | 6                 | ✓                      | ✓              | ✓        |
| Startup        | ✓         | 4 (+Company)      | ✓                      | ✓              | ✓        |
| Investor       | ✓         | 3                 | ✓ (derived)            | ✓              | ✓        |
| VC Fund        | ✓         | 4                 | ✓ (derived)            | ✓              | ✓        |
| Founder        | ✓         | 3 (+Person)       | ✓ (derived)            | ✓              | ✓        |

| Object               | Unique ID | Core Fields Count | Has Business Relevance | Has Timestamps | Has Tags |
|----------------------|-----------|-------------------|------------------------|----------------|----------|
| Research Institution | ✓         | 3                 | ✓ (derived)            | ✓              | ✓        |
| Government Policy    | ✓         | 6                 | ✓                      | ✓              | ✓        |
| Repository           | ✓         | 6                 | ✓ (derived)            | ✓              | ✓        |
| AI Model             | ✓         | 6                 | ✓ (derived)            | ✓              | ✓        |
| Framework            | ✓         | 4                 | ✓ (derived)            | ✓              | ✓        |
| Developer Tool       | ✓         | 4                 | ✓ (derived)            | ✓              | ✓        |
| Social Post          | ✓         | 5                 | ✓ (derived)            | ✓              | ✓        |
| Event                | ✓         | 4                 | ✓                      | ✓              | ✓        |

## SECTION 6 — RELATIONSHIP MODEL

### 6.1 Why Relationships Matter

The platform’s core value over a news reader is that no object exists in isolation. A paper is linked to the university that produced it, the authors who wrote it, the company they later founded, the funding it attracted, the investors involved, and the market trend that resulted. Surfacing that chain is the “why it matters” answer the platform exists to give.

### 6.2 Primary Relationship Chain

graph TD

```

A[Research Paper] --> B[University / Institution]
A --> C[Authors]
C --> D[Company / Startup]
D --> E[Funding Round]
E --> F[Investor / VC Fund]
F --> G[Market Trend]
D --> H[Open Source Repository]
D --> I[AI Model]
G --> J[News Coverage]
G --> K[Business Relevance: Viorant / Miraya]

```

### 6.3 Relationship Definitions

| Relationship                 | Description                                                                 |
|------------------------------|-----------------------------------------------------------------------------|
| Research Paper → Institution | Paper is affiliated with the institution(s) of its authors                  |
| Research Paper → Authors     | Many-to-many; authors may span institutions                                 |
| Authors → Company            | An author may found or join a company (tracked as a career-transition edge) |

| Relationship                     | Description                                                                                               |
|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| Company → Funding Round          | One-to-many; a company has a funding history over time                                                    |
| Funding Round → Investor/Fund    | Many-to-many; a round has one or more investors                                                           |
| Company → Open Source Repository | A company may publish/maintain repos                                                                      |
| Company → AI Model               | A company/lab may release models                                                                          |
| Company/Model/Research → News    | News items reference and contextualize any of the above                                                   |
| Any Object → Market Trend        | Aggregated signal across many objects forms a detectable trend (tag cluster + score velocity)             |
| Any Object → Business Relevance  | Every object can be evaluated against Viorant/Miraya focus areas independently of its other relationships |
| Investor → Company (Portfolio)   | An investor/fund holds a portfolio of companies, distinct from a single funding round                     |
| Policy → Company/Sector          | A government policy can name or affect specific companies or entire sectors                               |

## 6.4 Cross-Module Linking Rule

Any object detail page must render a **Related Objects rail** built from this relationship model — this is a mandatory IA requirement, not an optional enhancement, because it is the mechanism by which “why does it matter” gets answered without the user having to manually search.

---

## SECTION 7 — USER FLOWS

Each flow is defined by Trigger → Steps → Decision Points → Exit Points.

### 7.1 Morning Brief

- **Trigger:** User opens the platform for the first time in a day
- **Steps:** Dashboard loads → Top Critical/High priority items across all modules shown → User scans headline cards → User opens items of interest
- **Decision Points:** Read now vs. Add to Reading List vs. Dismiss (mark reviewed)
- **Exit Points:** All top items reviewed → Dashboard marked “caught up” for the day; or user jumps into a Deep Research flow from an item

### 7.2 Research Workflow

- **Trigger:** User clicks into Research module or a paper from Dashboard

- **Steps:** Browse/filter papers → Open Paper Detail → Read AI summary → View Related Objects (authors, institution, related companies) → Decide value
- **Decision Points:** Bookmark vs. Add to Reading List vs. Note vs. Ignore
- **Exit Points:** Paper saved to a Collection; or user pivots to the related Company/Institution page

### 7.3 Funding Discovery

- **Trigger:** User opens Funding module (daily habit or Dashboard alert)
- **Steps:** View feed sorted by newest/highest funding → Filter by sector/stage → Open Funding Round Detail → View investors and company profile
- **Decision Points:** Mark as Competitor vs. Mark as Opportunity vs. Just Track
- **Exit Points:** Company added to a Watchlist Collection; or user pivots to Investor Detail

### 7.4 Investor Tracking

- **Trigger:** User opens VC Intelligence or clicks an investor from a Funding Round
- **Steps:** View Investor Detail → Review portfolio and thesis tags → Compare to prior known activity
- **Decision Points:** Follow Investor (adds to personalization) vs. one-time view
- **Exit Points:** Investor added to Followed list, feeds future personalized surfacing

### 7.5 Research Paper Analysis

- **Trigger:** User wants to deeply evaluate one paper's implications
- **Steps:** Open Paper Detail → Read summary and metadata → Review related Company/Model links → Add annotation/Note → Assess Business Relevance
- **Decision Points:** Is this relevant to Viorant/Miraya? → Tag accordingly
- **Exit Points:** Paper tagged Business-Relevant and surfaced in Business Intelligence module

### 7.6 Bookmark Intelligence (Curation)

- **Trigger:** User finds a valuable item anywhere in the platform
- **Steps:** Click Bookmark → Optionally assign Collection/Tag → Optionally add Note
- **Decision Points:** New Collection vs. Existing Collection
- **Exit Points:** Item appears in Saved Intelligence → Bookmarks

### 7.7 Search Workflow

- **Trigger:** User has a specific question or recalls a past item
- **Steps:** Enter query into Global Search → System returns cross-module ranked results → User applies filters if needed → Opens result
- **Decision Points:** Keyword vs. natural language phrasing; refine with filters if too broad
- **Exit Points:** User reaches the object detail page or saves the search for reuse

7.8 Business Relevance Workflow

- **Trigger:** Any object is ingested or manually reviewed
- **Steps:** System evaluates object against Business Relevance rules (Section 12) → Assigns score and entity tags → High-relevance items surfaced on Dashboard and Business Intelligence module
- **Decision Points:** Auto-tagged vs. user manually overrides/confirms relevance
- **Exit Points:** Object appears in Business Intelligence feed; may trigger an Alert (Section 14)

7.9 Government Policy Workflow

- **Trigger:** New policy ingested or user browses Government module
- **Steps:** View policy feed → Filter by jurisdiction/impact area → Open Policy Detail → Assess relevance
- **Decision Points:** Track for compliance vs. informational only
- **Exit Points:** Policy bookmarked/noted; may feed a Custom Alert for that jurisdiction

7.10 Model Release Workflow

- **Trigger:** New model release ingested or user browses Models & Releases
- **Steps:** View release feed → Compare benchmark deltas → Open Model Detail → Review associated paper/lab
- **Decision Points:** Evaluate for internal use vs. track competitively
- **Exit Points:** Model bookmarked to a Collection (e.g., “Models to Evaluate”)

7.11 Saved Intelligence Workflow

- **Trigger:** User opens Saved Intelligence to review prior curation
- **Steps:** Choose sub-area (Bookmarks/Reading List/Notes/Collections) → Browse/filter/search within saved items → Reopen an object or edit a note
- **Decision Points:** Archive vs. keep vs. escalate to Business Intelligence
- **Exit Points:** Item archived, or reopened for continued research (loops back into Research Workflow)

SECTION 8 — SEARCH ARCHITECTURE

8.1 Search Modes

| Mode                    | Description                                                                                                  | Availability      |
|-------------------------|--------------------------------------------------------------------------------------------------------------|-------------------|
| Keyword Search          | Exact/fuzzy term match across title, summary, tags                                                           | MVP               |
| Natural Language Search | Query interpreted by Gemini into structured filters + keyword search (“seed rounds in agentic AI last week”) | MVP (AI-assisted) |
| Advanced                | Manual filter panel combinable with any search mode                                                          | MVP               |

| Mode                       | Description                                                                               | Availability           |
|----------------------------|-------------------------------------------------------------------------------------------|------------------------|
| Filters                    |                                                                                           |                        |
| AI Search<br>(answer mode) | Gemini synthesizes an answer from matching objects, with citations back to source objects | Phase 2                |
| Semantic Search            | Vector-embedding similarity search, independent of exact keyword match                    | Future<br>(Section 16) |

## 8.2 Search Fields

Search indexes the following fields across all object types: title/name, summary, tags[] (topic/technology/region), source, authors/entities mentioned, category. Object-specific fields (e.g., investor\_name on Funding, institution on Research Paper) are included in the index for that object type.

## 8.3 Indexing Strategy

| Layer                 | Approach                                                                                                                                                                                                                                   |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MVP                   | Relational database full-text search (e.g., Postgres tsvector/GIN index) — zero additional infrastructure cost                                                                                                                             |
| Query-time enrichment | Natural language queries are pre-processed by Gemini into a structured filter object (date range, object type, tags) before hitting the index                                                                                              |
| Future                | Vector index (e.g., pgvector extension on the same Postgres instance) storing embeddings per object for semantic search — chosen specifically because it avoids standing up a separate vector database, preserving the ₹0 budget principle |

## 8.4 Filtering Within Search

Search results respect the same filter taxonomy as module feeds (Section 9), applied as post-filters on the ranked result set. Users can scope search to one module or search globally (default).

## 8.5 Ranking of Search Results

Search results are ranked by a blend of **text relevance score** (match strength) and **Intelligence Score** (Section 11), so a strong keyword match on a low-credibility, stale item does not outrank a moderate match on a high-priority, fresh item. Business Relevance acts as a boost multiplier, not a hard filter, so relevant-but-not-business-tagged results are never hidden.

## 8.6 Search Suggestions

| Suggestion Type     | Trigger                                                    |
|---------------------|------------------------------------------------------------|
| Recent Searches     | Shown on search box focus, before typing                   |
| Trending Queries    | Based on what's spiking across ingested content this week  |
| Entity Autocomplete | As user types, suggest matching Company/Person/Institution |

| Suggestion Type        | Trigger names                                        |
|------------------------|------------------------------------------------------|
| Saved Search Shortcuts | User's saved searches surfaced as quick-access chips |

## SECTION 9 — FILTER ARCHITECTURE

### 9.1 Filter Taxonomy

Filters are grouped into five families and applied consistently, though not every module exposes every filter (a module only shows filters relevant to its object type).

| Family                    | Filters                                                          |
|---------------------------|------------------------------------------------------------------|
| <b>Temporal</b>           | Date, Date Range, Recency (Today/Week/Month)                     |
| <b>Geographic/Org</b>     | Country, Region, Company, Startup, Investor, Institution, Author |
| <b>Categorical</b>        | Technology, Model Type, Research Category, Source, Funding Stage |
| <b>Status</b>             | Status, Saved, Read, Unread, Bookmarked                          |
| <b>Priority/Relevance</b> | Priority, Business, Open Source, Popularity                      |

### 9.2 Filter-to-Module Applicability Matrix

| Filter            | News | Models | Funding | VC | Research | Universities | Government | Open Source | Dev Ecosystem | Social | Business Intel |
|-------------------|------|--------|---------|----|----------|--------------|------------|-------------|---------------|--------|----------------|
| Date              | ✓    | ✓      | ✓       | ✓  | ✓        | ✓            | ✓          | ✓           | ✓             | ✓      | ✓              |
| Country/Region    | ✓    | —      | ✓       | ✓  | —        | ✓            | ✓          | —           | —             | —      | ✓              |
| Technology        | ✓    | ✓      | ✓       | ✓  | ✓        | —            | —          | ✓           | ✓             | ✓      | ✓              |
| Company           | ✓    | ✓      | ✓       | —  | —        | —            | ✓          | ✓           | ✓             | ✓      | ✓              |
| Startup           | —    | —      | ✓       | ✓  | —        | —            | —          | —           | —             | —      | ✓              |
| Investor          | —    | —      | ✓       | ✓  | —        | —            | —          | —           | —             | —      | —              |
| Funding Stage     | —    | —      | ✓       | ✓  | —        | —            | —          | —           | —             | —      | ✓              |
| Research Category | —    | —      | —       | —  | ✓        | ✓            | —          | —           | —             | —      | —              |
| Source            | ✓    | ✓      | ✓       | —  | ✓        | —            | ✓          | ✓           | ✓             | ✓      | —              |
| Priority          | ✓    | ✓      | ✓       | ✓  | ✓        | ✓            | ✓          | ✓           | ✓             | ✓      | ✓              |
| Business          | ✓    | ✓      | ✓       | ✓  | ✓        | ✓            | ✓          | ✓           | ✓             | ✓      | ✓              |
| Open Source       | —    | —      | —       | —  | —        | —            | —          | ✓           | ✓             | —      | —              |
| Popularity        | ✓    | ✓      | —       | —  | ✓        | —            | —          | ✓           | ✓             | ✓      | —              |



| Filter                       | News | Models | Funding | VC | Research | Universities | Government | Open Source | Dev Ecosystem | Social | Business Intel |
|------------------------------|------|--------|---------|----|----------|--------------|------------|-------------|---------------|--------|----------------|
| Author                       | —    | —      | —       | —  | ✓        | ✓            | —          | —           | —             | ✓      | —              |
| Institution                  | —    | —      | —       | —  | ✓        | ✓            | —          | —           | —             | —      | —              |
| Model Type                   | —    | ✓      | —       | —  | —        | —            | —          | —           | —             | —      | —              |
| Status                       | ✓    | ✓      | ✓       | ✓  | ✓        | ✓            | ✓          | ✓           | ✓             | ✓      | ✓              |
| Saved/Read/Unread/Bookmarked | ✓    | ✓      | ✓       | ✓  | ✓        | ✓            | ✓          | ✓           | ✓             | ✓      | ✓              |

### 9.3 Filter Interaction Rules

- Filters within the same family are **OR**-combined (e.g., Country = India OR USA).
- Filters across families are **AND**-combined (e.g., Country = India AND Technology = LLM).
- Active filters persist per-module per-session and are shown as removable chips above the feed.
- “Saved Filter Sets” allow a user to name and reuse a combination (feeds into Personalization, Section 15).

## SECTION 10 — SORTING ARCHITECTURE

### 10.1 Sort Options

| Sort               | Logic                                                                                       | Applicable Modules                |
|--------------------|---------------------------------------------------------------------------------------------|-----------------------------------|
| Newest             | published_at descending                                                                     | All                               |
| Most Important     | Intelligence Score descending                                                               | All                               |
| Trending           | Score velocity (rate of engagement/mentions increase over trailing 48-72h)                  | News, Social, Open Source, Models |
| Highest Funding    | amount descending                                                                           | Funding                           |
| Most Credible      | source_credibility descending                                                               | News, Research                    |
| Business Relevance | Business Relevance Score descending                                                         | All                               |
| Most Discussed     | Community Engagement Score descending                                                       | News, Social, Open Source         |
| AI Score           | Composite Intelligence Score (alias of Most Important, exposed distinctly for transparency) | All                               |

## 10.2 Default Sort Per Module

| Module                | Default Sort                         |
|-----------------------|--------------------------------------|
| Dashboard             | Most Important (Critical/High first) |
| AI News               | Most Important                       |
| Models & Releases     | Newest                               |
| Funding               | Highest Funding (within last 7 days) |
| VC Intelligence       | Most Active                          |
| Research              | Most Important                       |
| Universities          | Most Active                          |
| Government            | Most Important                       |
| Open Source           | Trending                             |
| Developer Ecosystem   | Newest                               |
| Social Intelligence   | Trending                             |
| Business Intelligence | Business Relevance                   |

## 10.3 Sort/Filter Independence

Sorting is always applied **after** filtering, and changing sort never resets active filters — this is a mandatory UX/IA contract to prevent the frustration of losing a filtered view when trying a different sort order.

---

# SECTION 11 — INTELLIGENCE SCORING

## 11.1 Purpose

Every object receives a computed **Intelligence Score (0-100)** at ingestion time, recalculated periodically (e.g., daily) to account for freshness decay and new relationship signals (e.g., a paper gains score if it later gets cited by a well-funded startup).

## 11.2 Scoring Dimensions

| Dimension            | What it Measures                           | Applies Most Strongly To            |
|----------------------|--------------------------------------------|-------------------------------------|
| Credibility          | Trustworthiness/authority of the source    | News, Research, Government          |
| Freshness            | Recency, with decay curve over time        | All                                 |
| Business Impact      | Relevance to Viorant/Miraya focus areas    | All (via Business Relevance System) |
| Community Engagement | Discussion volume, stars, shares, comments | Social, Open Source, News           |
| Funding              | Size/significance of capital event         | Funding, VC                         |

| Dimension            | What it Measures                                     | Applies Most Strongly To                   |
|----------------------|------------------------------------------------------|--------------------------------------------|
| Research Importance  | Citation velocity, institution reputation, novelty   | Research, Universities                     |
| Strategic Value      | Long-term significance beyond immediate news cycle   | All (editorial/AI judgment)                |
| Technical Innovation | Degree of technical novelty or capability leap       | Models, Open Source, Developer Ecosystem   |
| Market Impact        | Potential to shift a market or competitive landscape | Funding, Business Intelligence, Government |

### 11.3 Composite Score Formula (conceptual, not implementation)

Intelligence Score =  
 (Credibility × W1) +  
 (Freshness × W2) +  
 (Business Impact × W3) +  
 (Community Engagement × W4) +  
 (Category-Specific Dimension × W5)

Weights (W1-W5) are **module-specific presets** — e.g., Research weighs Research Importance and Credibility heavily while Social weighs Community Engagement and Freshness heavily — configured centrally so tuning doesn't require touching individual modules.

### 11.4 Priority Classification

| Class    | Score Range | Meaning                                | Dashboard Treatment                                                                                 |
|----------|-------------|----------------------------------------|-----------------------------------------------------------------------------------------------------|
| Critical | 90-100      | Immediate strategic significance       | Pinned to top of Dashboard, triggers alert                                                          |
| High     | 70-89       | Important, should be reviewed same day | Prominent in feed                                                                                   |
| Medium   | 40-69       | Worth seeing, not urgent               | Standard feed position                                                                              |
| Low      | 15-39       | Marginal relevance                     | Collapsed/below fold by default                                                                     |
| Noise    | 0-14        | Low value                              | Filtered out of default views, retained in database, visible only via explicit "show all" or search |

### 11.5 Scoring Governance Table

| Rule          | Description                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| Recalculation | Freshness component decays continuously; other components recalculate on new relationship data (e.g., citation, funding follow-on) |
| Transparency  | Detail pages show a score breakdown by dimension, not just the final number, so the user can trust/audit it                        |

| Rule            | Description                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------|
| Manual Override | User can manually adjust Business Impact/Priority for their own view without affecting the global score |

## SECTION 12 — BUSINESS RELEVANCE SYSTEM

### 12.1 Purpose

This is the system that answers “does it affect my businesses?” — the single most differentiating capability versus a generic news/research aggregator.

### 12.2 Business Entities Tracked

| Entity             | Description                                                                          |
|--------------------|--------------------------------------------------------------------------------------|
| Viorant            | Primary business entity — focus areas configured in Settings                         |
| Miraya             | Secondary business entity — focus areas configured in Settings                       |
| Personal Investing | Optional third lens for personal market-watching interests not tied to either entity |

### 12.3 Relevance Categories

| Category               | Meaning                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| Fundraising            | Item relevant to Viorant/Miraya’s own fundraising strategy (comparable rounds, active investors in the space) |
| Investments            | Item relevant to potential personal/business investment decisions                                             |
| Partnerships           | Item surfaces a potential partner (technology, distribution, research)                                        |
| Product Ideas          | Item suggests a feature, model, or capability worth incorporating                                             |
| Competitors            | Item involves a company competing directly or adjacently                                                      |
| Customer Opportunities | Item surfaces a potential customer or market segment                                                          |

### 12.4 Tagging Model

Each object can carry zero or more `relevant_entities[]` (Viorant / Miraya / Personal) and zero or more `relevance_category[]` (from 12.3 table), plus a free-text `relevance_reason` (AI-generated rationale, e.g., “New funding round in the same vertical as Viorant, by an investor Viorant has approached”).

### 12.5 Relevance Calculation

flowchart LR

```

    A[New Object Ingested] --> B{Match against Business Profile
    keywords/sectors/competitors}
    B -->|Match found| C[Assign relevant_entities + relevance_category]
    C --> D[Compute Business Impact sub-score]
```

```

D --> E[Feed into Intelligence Score W3]
B -->|No match| F[relevance_score = 0, no tags]
E --> G[Surface in Business Intelligence module + Dashboard]

```

The **Business Profile** (configured under Settings → Business Entities) stores keywords, sectors, named competitors, named target customers, and named target investors per entity. Matching is keyword/entity-based at MVP, upgraded to embedding-similarity matching once vector search (Section 16) ships, allowing conceptual relevance (“this is adjacent to our space”) rather than only exact-term relevance.

## 12.6 Business Relevance Score Weighting

| Match Type                           | Weight  |
|--------------------------------------|---------|
| Named competitor/entity direct match | Highest |
| Sector/keyword match                 | High    |
| Adjacent technology match            | Medium  |
| Weak/tangential thematic overlap     | Low     |

# SECTION 13 — PERSONAL KNOWLEDGE SYSTEM

## 13.1 Components

| Component       | Purpose                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Bookmarks       | Quick single-click save of any object, no organization required                                                                         |
| Reading List    | Queue of items intended for deeper reading, separate from casual bookmarks                                                              |
| Collections     | User-named groups of objects (e.g., “Competitors to Watch,” “Models to Evaluate”) — can mix object types                                |
| Folders         | Optional nesting of Collections for heavier organization (e.g., Collections grouped under a “Fundraising” folder)                       |
| Tags            | Free-form user tags, distinct from system topic_tags, fully user-controlled                                                             |
| Notes           | Freeform annotations attached to any object, or standalone notes not tied to an object                                                  |
| Highlights      | Selected excerpts from summaries the user marks as key (paraphrase-safe, no raw copyrighted text stored beyond fair-use snippet length) |
| Saved Searches  | Named, reusable search+filter combinations                                                                                              |
| History         | Automatic log of every object viewed, timestamped                                                                                       |
| Recent Activity | Rolling view of the user’s last N actions (bookmarks, notes, searches) across the platform                                              |

## 13.2 Structural Relationship

### Saved Intelligence

- Bookmarks (flat list, quick-save)
- Reading List (queue, ordered)
- Collections (named groups)
  - Folders (optional grouping of Collections)
- Notes (attached to object OR standalone)
- Saved Searches
- History / Recent Activity (auto-generated, read-only)

## 13.3 Object-to-Knowledge-System Linking Rule

Every object detail page across every module exposes the same four actions consistently: **Bookmark, Add to Reading List, Add to Collection, Add Note** — this consistency is mandatory so the user never has to relearn curation mechanics per module.

---

## SECTION 14 — NOTIFICATION ARCHITECTURE

### 14.1 Notification Types

| Type                 | Cadence                      | Content                                                                         |
|----------------------|------------------------------|---------------------------------------------------------------------------------|
| Daily Brief          | Daily (user-configured time) | Top Critical/High priority items across all modules from the last 24h           |
| Weekly Summary       | Weekly                       | Trend overview, top items by category, notable business-relevant items          |
| Monthly Report       | Monthly                      | Longer-arc trend analysis, funding/research volume comparisons month-over-month |
| Breaking News        | Real-time (as-detected)      | Single Critical-priority News item                                              |
| Funding Alerts       | Real-time/batched            | New funding round matching saved filters or business relevance                  |
| Research Alerts      | Real-time/batched            | New paper matching followed authors/institutions/topics                         |
| Government Alerts    | Real-time/batched            | New policy matching followed jurisdictions                                      |
| Model Release Alerts | Real-time/batched            | New model release from followed labs                                            |
| Custom Alerts        | User-defined                 | User builds a rule (module + filter combination + threshold)                    |

### 14.2 Notification Delivery Channels (architecture-level, not implementation)

| Channel | MVP Availability |
|---------|------------------|
|---------|------------------|

| Channel                    | MVP Availability                               |
|----------------------------|------------------------------------------------|
| In-app notification center | Yes                                            |
| Email digest               | Yes (low-cost, scheduled batch)                |
| Push notification          | Future (requires mobile app or PWA push infra) |

### 14.3 Notification Rule Structure

Alert = Module(s) + Filter Criteria + Score Threshold + Frequency (real-time / daily digest / weekly digest)

This single structure powers every notification type in the table above — Daily Brief and Custom Alerts are the same underlying object with different default configurations, keeping the notification system to one engine rather than N bespoke ones.

### 14.4 Notification Governance

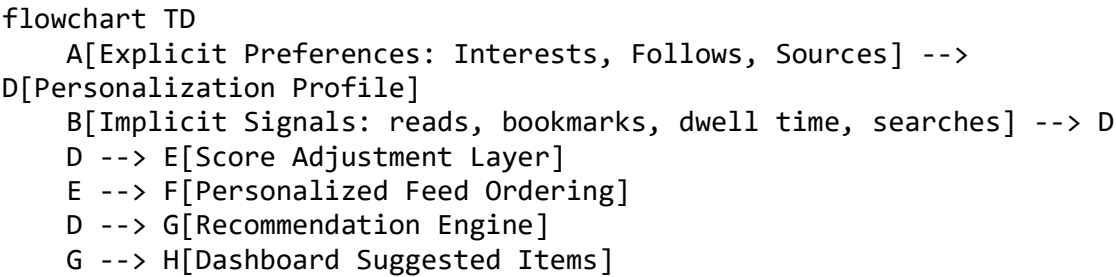
| Rule                      | Description                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------|
| No duplicate alerts       | An object that triggers multiple matching rules is only delivered once per channel per day, with the triggering rules listed |
| User-controlled volume    | All non-critical notification types can be muted or re-scheduled independently in Settings                                   |
| Critical always delivered | Critical-priority Business-Relevant items bypass digest batching and notify immediately, regardless of other settings        |

## SECTION 15 — PERSONALIZATION

### 15.1 Personalization Inputs

| Input                 | Description                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------|
| Interests             | User-selected topics/technologies weighted higher in scoring and sort                         |
| Favorite Companies    | Followed companies surfaced prominently and alerted on                                        |
| Favorite Investors    | Followed investors/funds surfaced prominently and alerted on                                  |
| Favorite Technologies | Followed technology tags (e.g., “agentic AI,” “robotics”)                                     |
| Preferred Sources     | Sources weighted up in credibility/visibility                                                 |
| Ignored Sources       | Sources suppressed from default feeds (not deleted, still searchable)                         |
| Learning Preferences  | Implicit signal from read/bookmark/dwell-time behavior, feeding a lightweight relevance model |
| Recommendation Engine | Surfaces items the user didn’t explicitly follow but is likely to value, based on the above   |

15.2 Personalization Data Flow



15.3 Personalization vs. Global Scoring

Personalization **re-ranks** within a module/feed; it never overrides the objective Intelligence Score used for cross-user consistency (important once multi-user ships — the underlying score stays comparable across users, only presentation order changes). This separation is a mandatory architectural rule.

15.4 Personalization Configuration Surface

All personalization inputs live under **Settings → Personalization**, with Favorite Companies/Investors/Technologies also settable inline via a “Follow” action on any relevant object detail page — a dual-entry-point pattern used consistently (configure centrally, or follow contextually).

SECTION 16 — FUTURE MODULES

16.1 Roadmap Table

| Future Module   | Description                                                                                                        | Depends On                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| AI Chat         | Conversational interface over the entire intelligence database (“what happened in agentic AI funding this month?”) | Search Architecture (semantic layer), Gemini API                  |
| Voice Assistant | Voice-driven brief/query interface                                                                                 | AI Chat                                                           |
| Multi-user      | Team accounts, shared and private views                                                                            | User-scoped data model (already designed in from V1, Section 1.6) |
| Team Workspace  | Shared Collections, shared Notes, assignment of items to teammates                                                 | Multi-user                                                        |
| Knowledge Graph | Visual, explorable graph rendering of the Relationship Model (Section 6)                                           | Relationship Model maturity, sufficient object volume             |
| Timeline View   | Chronological visualization of an entity’s history (e.g., a company’s full                                         | Relationship Model, historical database depth                     |



| Future Module            | Description                                                                    | Depends On                                                |
|--------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------|
|                          | funding/news/research timeline)                                                |                                                           |
| Trend Prediction         | Forecasting emerging topics/technologies based on score velocity patterns      | Historical database (12+ months), scoring engine maturity |
| Competitive Intelligence | Structured competitor-tracking dashboards beyond basic tagging                 | Business Relevance System maturity                        |
| Market Forecasting       | Sector-level sizing and trajectory estimates                                   | Trend Prediction, external market data sources            |
| Portfolio Tracking       | If user becomes an investor, track own portfolio companies within the platform | VC Intelligence module, multi-entity data model           |
| Investor CRM             | Relationship/contact management layered onto VC Intelligence                   | VC Intelligence, Multi-user (for shared CRM use)          |

### 16.2 Sequencing Logic

Future modules are ordered by dependency, not by desirability — e.g., Knowledge Graph and Trend Prediction are gated on data volume/maturity regardless of demand, while AI Chat is gated on the semantic search layer already planned for Phase 2. This dependency-first sequencing is documented so future roadmap decisions don't attempt to build a module before its prerequisite architecture exists.

## SECTION 17 — MASTER INFORMATION HIERARCHY

### 17.1 Universal Hierarchy Diagram

```
graph TD
  A[Platform] --> B[Module]
  B --> C[Category]
  C --> D[Subcategory]
  D --> E[Object]
  E --> F[Detail Page]
  F --> G[Related Objects]
  F --> H[Actions]
  G --> E
```

### 17.2 Worked Example — Research Module

```
Platform
├── Research (Module)
│   ├── AI Safety (Category)
│   │   ├── Alignment Techniques (Subcategory)
│   │   │   └── "Paper: Scalable Oversight via Debate" (Object)
```

└─ Detail Page  
    └─ Related Objects: Authors, Institution, Cited-by  
Papers, Spinout Company  
    └─ Actions: Bookmark, Add to Reading List, Note,  
Mark Business-Relevant

### 17.3 Why This Hierarchy Is Load-Bearing

This exact seven-level structure is what every module, page hierarchy (Section 3), and object model (Section 5) is built against. A UI/UX designer can design **one** detail-page template, **one** feed-page template, and **one** related-objects component, and apply them across all twelve modules — because every module obeys this same hierarchy. This is the single biggest cost/time saver enabled by the IA, directly serving the near-₹0 budget constraint.

---

## SECTION 18 — SITEMAP

### 18.1 Full Sitemap (all levels)

graph TD

```
Root[Platform Root] --> Dash[Dashboard]
Root --> News[AI News]
Root --> Models[Models & Releases]
Root --> Funding[Funding]
Root --> VC[VC Intelligence]
Root --> Research[Research]
Root --> Univ[Universities]
Root --> Gov[Government]
Root --> OSS[Open Source]
Root --> Dev[Developer Ecosystem]
Root --> Social[Social Intelligence]
Root --> BI[Business Intelligence]
Root --> Saved[Saved Intelligence]
Root --> Search[Global Search]
Root --> Settings[Settings]
Root --> Profile[Profile]
Root --> Admin[Admin]

News --> NewsD[News Detail]
Models --> ModelsD[Model Detail]
Funding --> FundD[Funding Round Detail]
VC --> Investors[Investor List]
VC --> Funds[Fund List]
Investors --> InvD[Investor Detail]
Funds --> FundsD[Fund Detail]
Research --> ResD[Paper Detail]
Univ --> UnivD[Institution Detail]
```

Gov --> GovD[Policy Detail]  
OSS --> OSSD[Repository Detail]  
Dev --> DevD[Tool Detail]  
Social --> SocD[Post Detail]  
BI --> Companies[Company List]  
BI --> Reports[Report List]  
Companies --> CompD[Company Detail]  
Reports --> RepD[Report Detail]  
  
Saved --> Bookmarks  
Saved --> ReadList[Reading List]  
Saved --> Notes  
Saved --> Collections  
Saved --> SavedSearch[Saved Searches]  
  
Settings --> SetSources[Sources]  
Settings --> SetAlerts[Alerts]  
Settings --> SetPers[Personalization]  
Settings --> SetBiz[Business Entities]  
  
Admin --> AdSources[Source Management]  
Admin --> AdIngest[Ingestion Health]  
Admin --> AdUsers[User Management]

---

## SECTION 19 — NAVIGATION DIAGRAM

### 19.1 Primary Navigation Flow

flowchart LR

Dashboard -->|click module| ModuleFeed[Module Feed]  
ModuleFeed -->|apply filter/sort| ModuleFeed  
ModuleFeed -->|click item| DetailPage[Object Detail Page]  
DetailPage -->|click related object| DetailPage  
DetailPage -->|bookmark/note/collection| Saved[Saved Intelligence]  
Dashboard -->|global search| SearchResults[Search Results]  
SearchResults -->|click item| DetailPage  
Saved -->|reopen item| DetailPage  
DetailPage -->|back| ModuleFeed  
ModuleFeed -->|back| Dashboard

### 19.2 Navigation State Persistence Rules

| Rule                          | Behavior                                               |
|-------------------------------|--------------------------------------------------------|
| Back from Detail → Feed       | Restores prior scroll position and active filters/sort |
| Back from Feed →<br>Dashboard | Dashboard reloads fresh top items (not cached state)   |

| Rule                                 | Behavior                                                                                                      |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Search → Detail → Back               | Returns to search results, preserving query and filters                                                       |
| Cross-module jump via Related Object | Pushes a new navigation entry (not a replace), so back-button history remains intact across module boundaries |

## SECTION 20 — FINAL INFORMATION ARCHITECTURE SUMMARY

### 20.1 Overall Architecture Philosophy

This IA is built on one governing idea: **twelve content domains, one structural skeleton**. Every module — regardless of whether it tracks a research paper or a funding round — obeys the same object envelope (Section 5.1), the same seven-level hierarchy (Section 17), the same filter/sort/scoring engine (Sections 9-11), and the same curation and navigation mechanics (Sections 13, 19). This uniformity is what makes a near-₹0, single-developer-feasible MVP possible while still supporting Bloomberg-Terminal-grade depth over time.

### 20.2 Scalability

| Dimension      | How the IA supports it                                                                                                                         |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Content volume | Object model and scoring engine are domain-agnostic; adding volume within existing modules requires no structural change                       |
| New modules    | Any new content domain that fits the Common Object Envelope (Section 5.1) can be added as a thirteenth module without touching the core system |
| Multi-user     | Every object and preference is already scoped to a user_id conceptually (Section 1.6); scaling to teams is a permissions layer, not a redesign |
| Search scale   | Relational full-text search at MVP, upgrading to vector search on the same store, avoiding a re-platforming event                              |

### 20.3 Extensibility

New filters, tags, and scoring dimensions can be added to the shared taxonomy (Sections 9, 11) without per-module rework, because modules consume the taxonomy rather than defining their own. New relationship types extend the graph (Section 6) additively.

### 20.4 Maintainability

A single detail-page template, single feed template, and single related-objects component (Section 17.3) mean one engineering change (e.g., improving how scores are displayed) automatically applies everywhere, rather than requiring twelve separate edits.

### 20.5 Future Readiness

The Future Modules (Section 16) were deliberately sequenced by architectural dependency rather than desirability, so the roadmap never requires retrofitting a foundational

capability (e.g., vector search, multi-user scoping) after the fact — those foundations are already present in the V1 design even where the feature built on top of them is not.

## 20.6 Recommended MVP Scope

| Included in MVP                                             | Rationale                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| All 11 content modules (feed + detail + basic filters/sort) | Core value proposition requires full domain coverage, even if shallow per-module at first   |
| Common Object Envelope + all object types                   | Needed from day one so relationships and search work correctly later without data migration |
| Keyword + natural-language search (Gemini-assisted)         | High value, low infra cost                                                                  |
| Intelligence Scoring (rule-based, not ML)                   | Deliverable with deterministic rules; no training infrastructure required                   |
| Business Relevance System (keyword/entity matching)         | Core differentiator, cheap to implement at MVP fidelity                                     |
| Bookmarks, Reading List, Notes, Collections                 | Core retention loop                                                                         |
| Daily Brief + Email digest                                  | Cheapest notification channel with highest habit-forming value                              |
| Basic Personalization (explicit follows only)               | Implicit learning deferred to Phase 2                                                       |

## 20.7 Recommended Phase 2

| Phase 2 Addition                               | Rationale                                                     |
|------------------------------------------------|---------------------------------------------------------------|
| Vector/semantic search                         | Materially improves discovery once object volume justifies it |
| AI Search (answer mode)                        | Builds directly on semantic search                            |
| Implicit personalization (behavioral learning) | Requires several months of usage history to be meaningful     |
| Push notifications                             | Requires mobile/PWA investment                                |
| Knowledge Graph visualization                  | Requires relationship density from months of ingestion        |
| Custom Alerts (full rule builder)              | Refines the MVP's fixed alert types into user-defined rules   |

## 20.8 Recommended Phase 3

| Phase 3 Addition                      | Rationale                                                 |
|---------------------------------------|-----------------------------------------------------------|
| Multi-user / Team Workspace           | Only valuable once single-user model is proven and stable |
| Trend Prediction / Market Forecasting | Requires 12+ months of historical scoring data            |

| Phase 3 Addition                  | Rationale                                                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| Investor CRM / Portfolio Tracking | Valuable only if user's own investing activity scales to warrant it                          |
| AI Chat / Voice Assistant         | Highest-effort future module; best built once semantic search and knowledge graph are mature |

## 20.9 Closing Statement

This document defines the complete structural skeleton of the platform — every module, object, relationship, flow, and governance rule required before any pixel is designed or line of code is written. Design and engineering teams can treat Sections 1-20 as the binding contract for what the platform contains and how its pieces relate; the next deliverable (UI/UX design) should apply visual and interaction design **within** this structure, not redefine it.

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*End of Deliverable 2 — Information Architecture Document.*